



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
TRIAL EXAMINATIONS 2005

FORM VI

MATHEMATICS EXTENSION 1

Examination date

Wednesday 10th August 2005

Time allowed

2 hours

Instructions

- All seven questions may be attempted.
- All seven questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

Checklist

- SGS booklets: 7 per boy. A total of 1000 booklets should be sufficient.
- Candidature: 117 boys.

Examiner

KWM

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

(a) Evaluate $\sum_{n=1}^4 n!$.

1

(b) Differentiate the following with respect to x .

(i) $y = \log_e(\sin x)$

1

(ii) $y = \cos^{-1} 3x$

1

(c) State the domain and range of the function $f(x) = 2 \cos^{-1} \frac{x}{3}$.

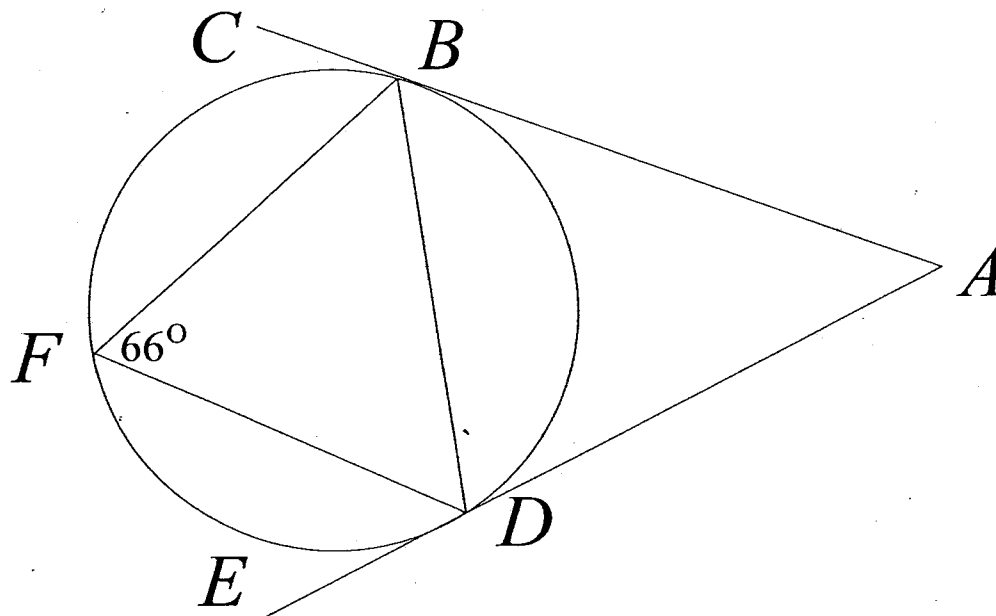
2

(d) Given the points $A(5, 1)$ and $B(-3, 6)$, find the co-ordinates of the point P that divides the interval AB externally in the ratio 3 : 4.

2

(e)

2



The diagram above shows the tangents AC and AE drawn to a circle. BF and DF are chords drawn from the points of contact at B and D respectively. Given that $\angle BFD = 66^\circ$, find $\angle BAD$ giving reasons for your answer.

(f) Use the substitution $u = 1 - x^2$ to evaluate the definite integral

3

$$\int_0^{\frac{\sqrt{3}}{2}} x\sqrt{1-x^2} dx.$$

QUESTION TWO (12 marks) Use a separate writing booklet.**Marks**

(a) Simplify $\frac{{}^nC_{r+1}}{{}^nC_r}$.

2

(b) Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^{12}$.

3

(c) A couple, purchasing a house, negotiates a \$300 000 mortgage to be repaid in equal monthly instalments over a period of 25 years. The interest on the loan is 7.2% per annum, compounded monthly. Let $\$A_n$ be the amount owing on the loan after n months, and $\$M$ the monthly repayment.

(i) Write down an expression for A_1 .

1

(ii) Hence show that $A_2 = 300\,000(1.006)^2 - 1.006M - M$.

1

(iii) Show that $A_n = 300\,000(1.006)^n - \frac{M(1.006^n - 1)}{0.006}$.

1

(iv) Find, to the nearest dollar, the monthly repayment M required to repay the loan over 25 years under the agreed terms.

1

(d) Find $\int_0^{\frac{\pi}{3}} \tan^2 x \, dx$.

3

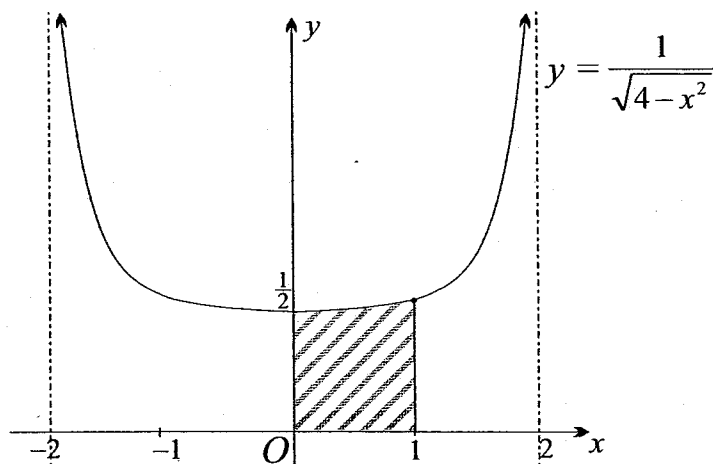
QUESTION THREE (12 marks) Use a separate writing booklet.

Marks

(a) Prove that $\frac{1 - \cos 2A}{\sin 2A} = \tan A$.

2

(b)



In the diagram above the curve $y = \frac{1}{\sqrt{4-x^2}}$ is sketched showing vertical asymptotes at $x = -2$ and $x = 2$. Find the exact area of the shaded region bounded by the curve, the line $x = 1$ and the co-ordinate axes.

2

(c) Let the equation $x^3 - 3x^2 - 4x + 12 = 0$ have roots α , β and γ .

(i) Find the value of $\alpha + \beta + \gamma$.

1

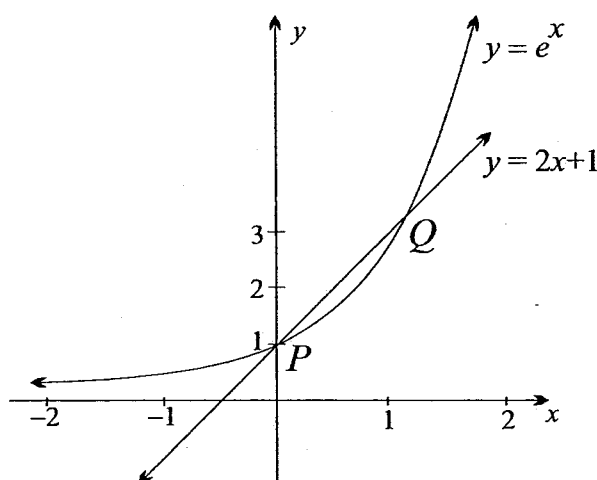
(ii) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

2

(iii) Given that two of its roots sum to zero, solve the equation.

2

(d)



The diagram above shows the curve $y = e^x$ and the line $y = 2x + 1$ intersecting at point $P(0, 1)$ and at another point Q . Use Newton's Method once, with initial approximation $x = 1$, to find a better approximation to the x co-ordinate of the point Q . Write your approximation correct to one decimal place.

3

Exam continues next page ...

QUESTION FOUR (12 marks) Use a separate writing booklet.**Marks**(a) Find $\cos^{-1}(-\frac{1}{2}) + \sin^{-1}(-\frac{\sqrt{3}}{2})$ in radians.**1**(b) Find the values of a and b that make the polynomial $P(x) = 2x^3 + ax^2 - 13x + b$ exactly divisible by $x^2 - x - 6$.**3**(c) (i) Express $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.**2**

(ii) Hence, or otherwise, find the general solution of the equation

2

$$\cos x - \sqrt{3}\sin x = 1.$$

(d) If the surrounding air temperature is 20°C , it takes 15 minutes for a cup of tea at a temperature of 80°C to cool to a temperature of 40°C . Given that T is the temperature in degrees Celsius of the tea after t minutes, then Newton's Law of cooling states that T satisfies the differential equation $\frac{dT}{dt} = k(T - 20)$.(i) Show that $T = 20 + Ae^{kt}$ is a solution of the differential equation.**1**(ii) Find the value of A , and show that $k = -\frac{\ln 3}{15}$.**2**

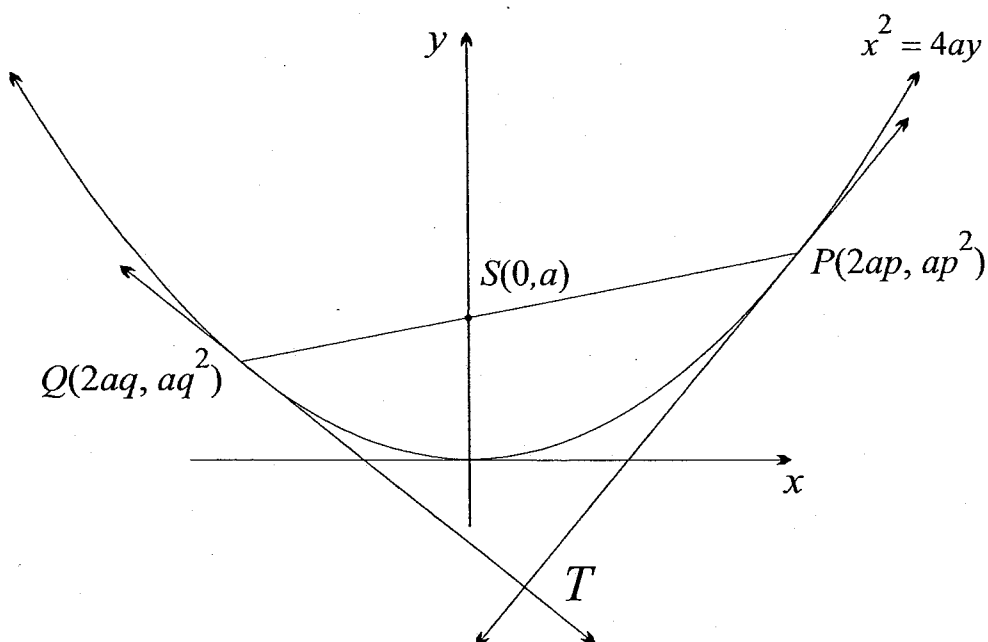
(iii) Find the temperature of the tea after 30 minutes.

1

QUESTION FIVE (12 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above a focal chord PQ intersects the parabola $x^2 = 4ay$ at points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$. The tangents to the parabola at point P and point Q intersect at T .

(i) Show that the equation of the tangent to the parabola at the point P is given by $y = px - ap^2$. 2

(ii) Show that $pq = -1$. 2

(iii) Show that the acute angle between the focal chord QP and the tangent TP to the parabola at P is given by $\tan^{-1} |q|$. 2

(b) A particle is moving in simple harmonic motion about the origin.

(i) Assuming that $\ddot{x} = -n^2x$, show that $v^2 = n^2(a^2 - x^2)$, where a is the amplitude. 2

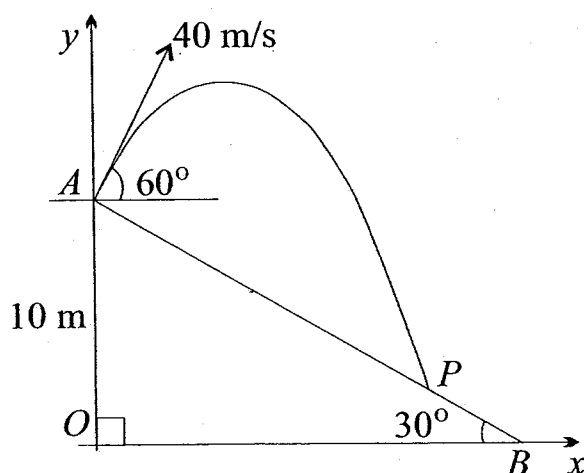
(ii) When the particle is 3 metres from the origin, its speed is 8 m/s, and when it is 4 metres from the origin its speed is 6 m/s. Find the period and amplitude of the motion. 3

(iii) Find the greatest acceleration of the particle. 1

QUESTION SIX (12 marks) Use a separate writing booklet.

Marks

(a)



The diagram above shows a plane inclined at 30° to the horizontal, meeting level ground at B . A ball is projected from a point A on the plane, 10 metres above the horizontal. The angle of projection is 60° to the horizontal and the initial speed of the ball is 40 m/s.

- (i) Take $g = 10 \text{ m/s}^2$, and show that the displacement equations of motion of the ball are given by

3

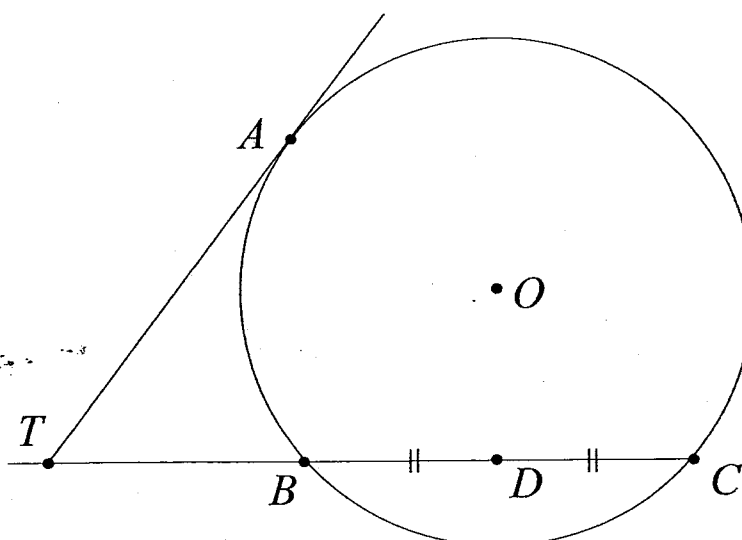
$$y = 20\sqrt{3}t - 5t^2 + 10 \quad \text{and}$$

$$x = 20t.$$

- (ii) Show that the ball hits the inclined plane at the point P after $t = \frac{16\sqrt{3}}{3}$ seconds.

3

(b)



In the diagram above, TA is a tangent and TBC is a secant drawn to a circle of centre O . Let the midpoint of the chord BC be D . Prove that $\angle AOT = \angle ADT$.

3

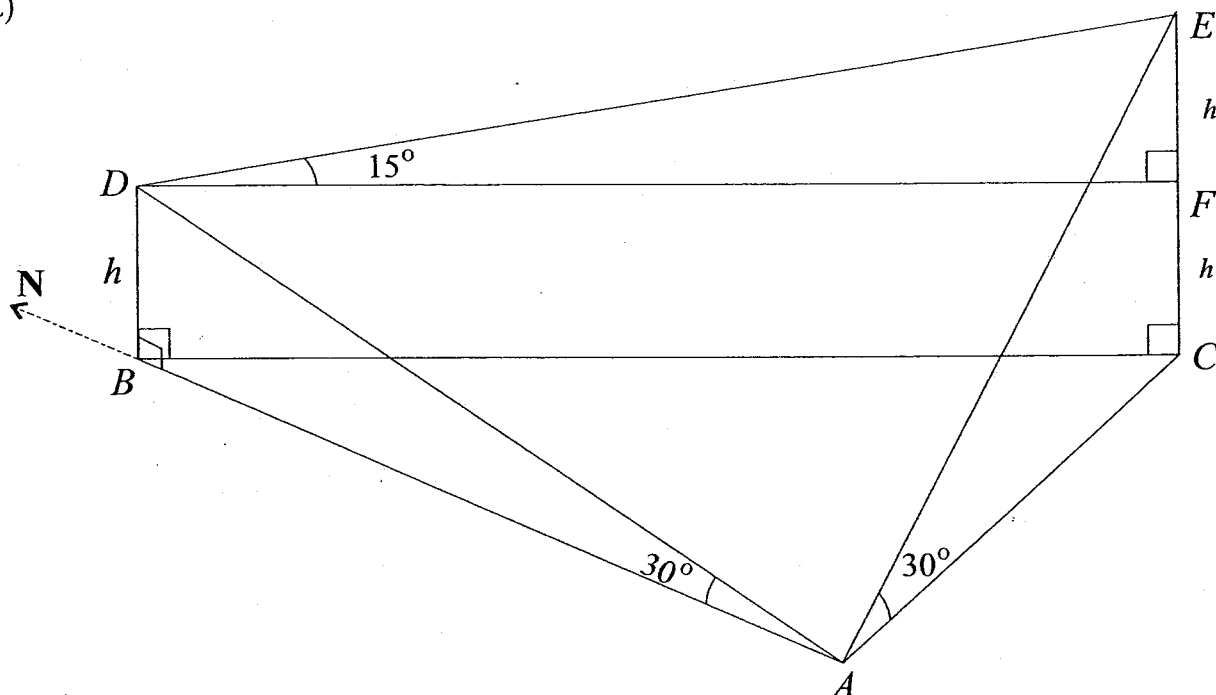
Exam continues overleaf ...

- (c) A rectangle is expanding in such a way that at all times it is twice as long as it is wide. If its area is increasing at a rate of $18 \text{ cm}^2/\text{s}$, find the rate at which its perimeter is increasing at the instant its width is 1 metre. 3

QUESTION SEVEN (12 marks) Use a separate writing booklet.

Marks

(a)



The diagram above shows two vertical towers BD and CE of heights h and $2h$ respectively, on a horizontal plane ABC . Point A is due south of point B , and the angles of elevation of the tops of the towers from A are both 30° . Given that the angle of elevation from D to E is 15° , find the bearing of the taller tower from point A correct to the nearest degree. 4

- (b) By considering the expansion of $(1+x)^{n-1}$, prove that: 4

$$\frac{7}{1} \binom{n-1}{0} + \frac{7^2}{2} \binom{n-1}{1} + \frac{7^3}{3} \binom{n-1}{2} + \cdots + \frac{7^n}{n} \binom{n-1}{n-1} = \frac{1}{n} (2^{3n} - 1).$$

- (c) Use induction, or otherwise, to prove that the sum of the products of all the pairs of different integers that can be formed from the first n positive integers is 4

$$\frac{n}{24} (n-1)(n+1)(3n+2).$$

END OF EXAMINATION

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

QUESTION 1

(a) $\sum_{n=1}^4 n! = 1! + 2! + 3! + 4!$
 $= 1 + 2 + 6 + 24$
 $= 33 \checkmark$

(b) (i) $y = \ln(\sin x)$
 $y' = \frac{1}{\sin x} \times \cos x$
 $y' = \cot x \checkmark$

(ii) $y = \cos^{-1} 3x$
 $y' = \frac{1}{\sqrt{1-9x^2}} \times 3$
 $y' = \frac{3}{\sqrt{1-9x^2}} \checkmark$

(c) $f(x) = 2 \cos^{-1} \frac{x}{3}$

Domain: $-1 \leq \frac{x}{3} \leq 1$
 so $-3 \leq x \leq 3 \checkmark$

Range: $0 \leq y \leq 2\pi \checkmark$

(d) A(5, 1), B(-3, 6)
 K: l = -3:4

$P \left(\frac{lx_1 + Ky_2}{1+K}, \frac{ly_1 + Kx_2}{1+K} \right) \checkmark$

$P \left(\frac{20+9}{1}, \frac{4-18}{1} \right)$

$P(29, -14) \checkmark$

(e) $\angle DBA = 66^\circ$ (The angle between a chord and tangent at the point of contact equals the angle drawn in the alternate segment.) $\checkmark$

Tangents drawn from an external point are equal, so $AB = AD$ and $\triangle ABD$ is isosceles.

$\angle BAD = \angle DBA = 66^\circ$

(base angles of an isosceles triangle are equal.)

so $\angle BAD = 180^\circ - 2 \times 66^\circ$
 $= 48^\circ \checkmark$

(angle sum of a triangle.)

(f) $\int_0^{\frac{\sqrt{2}}{2}} x\sqrt{1-x^2} dx$ $u = 1-x^2$
 $du = -2x dx$
 $-\frac{1}{2} du = x dx$

when $x=0$, $u=1 \checkmark$

when $x = \frac{\sqrt{2}}{2}$, $u = \frac{1}{2}$

$-\frac{1}{2} \int_1^{\frac{1}{2}} u^{\frac{1}{2}} du = \frac{1}{3} \left[u^{\frac{3}{2}} \right]_{\frac{1}{2}}^1$

$= \frac{1}{3} \left(1 - \frac{1}{8} \right)$

$= \frac{1}{3} \times \frac{7}{8}$

$= \frac{7}{24} \checkmark$

QUESTION 2

$$(a) \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n!}{(n-r-1)!(r+1)!} \times \frac{(n-r)!r!}{n!}$$

$$= \frac{n-r}{r+1} \checkmark$$

$$(b) \left(3x^2 + \frac{2}{x}\right)^{12}$$

General term $T_r = {}^{12}C_r (3x^2)^{12-r} \left(\frac{2}{x}\right)^r$

using rules of indices:

$$24 - 3r = 0$$

$$3r = 24$$

$$r = 8 \checkmark$$

$${}^{12}C_8 \times 3^4 \times 2^8 \checkmark$$

(c)

$$(i) A_1 = 300000 \times 1.006 - M = \left(\sqrt{3} - \frac{\pi}{3}\right) - (0)$$

$$(ii) A_2 = A_1 \times 1.006 - M = \sqrt{3} - \frac{\pi}{3} \checkmark$$

$$= 300000 (1.006)^2 - M (1.006) - M$$

$$(iii) A_n = 300000 (1.006)^n - M(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$

a geometric progression

$a = 1$ and $r = 1.006$.

$$S_n = \frac{a(r^n - 1)}{r - 1} \checkmark$$

$$S_n = \frac{(1.006)^n - 1}{0.006}$$

$$A_n = 300000 (1.006)^n - \frac{M (1.006^n - 1)}{0.006}$$

(iv) after the loan is repaid

$A_n = 0$, and $n = 300$

$$M = \frac{300000 (1.006)^{300} (0.006)}{(1.006)^{300} - 1}$$

$$M \div \$2159 \checkmark$$

$$(d) \int_0^{\frac{\pi}{3}} \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{3}} \sec^2 x - 1 \, dx \checkmark$$

$$= \left[\tan x - x \right]_0^{\frac{\pi}{3}} \checkmark$$

$$= \left(\sqrt{3} - \frac{\pi}{3}\right) - (0)$$

$$= \sqrt{3} - \frac{\pi}{3} \checkmark$$

(12)

QUESTION 3

$$\begin{aligned}
 (a) \quad LHS &= \frac{1 - \cos 2A}{\sin 2A} \\
 &= \frac{1 - (1 - 2\sin^2 A)}{2\sin A \cos A} \checkmark \\
 &= \frac{2\sin^2 A}{2\sin A \cos A} \\
 &= \frac{\sin A}{\cos A} \checkmark \\
 &= \tan A \\
 &= RHS
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \text{Area} &= \int_0^1 \frac{dx}{\sqrt{4-x^2}} \\
 &= \left[\sin^{-1} \frac{x}{2} \right]_0^1 \checkmark \\
 &= \frac{\pi}{6} - 0 \\
 &= \frac{\pi}{6} \text{ sq. units. } \checkmark
 \end{aligned}$$

$$(c) \quad x^3 - 3x^2 - 4x + 12 = 0$$

$$\begin{aligned}
 (i) \quad \alpha + \beta + \gamma &= -\frac{b}{a} \\
 &= 3 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \checkmark \\
 &= \frac{\frac{c}{a}}{-\frac{d}{a}} \\
 &= -\frac{c}{d} \\
 &= -\frac{1}{3} \checkmark
 \end{aligned}$$

$$(iii) \quad \text{Let the roots be } \alpha, -\alpha, \beta.$$

$$\text{then } \alpha - \alpha + \beta = 3$$

$$\text{and } \beta = 3. \checkmark$$

$$\text{now } \alpha(-\alpha)\beta = -12$$

$$-\alpha^2 = -4$$

$$\alpha = \pm 2$$

$$\text{the roots are } -2, 2 \text{ and } 3. \checkmark$$

$$\begin{aligned}
 (d) \quad y &= e^x \\
 y &= 2x+1
 \end{aligned}$$

The points of intersection correspond to the roots of the equation

$$e^x - 2x - 1 = 0$$

$$f(x) = e^x - 2x - 1. \checkmark$$

$$f'(x) = e^x - 2$$

$$\text{put } x_0 = 1 : f(1) = -0.282$$

$$f'(1) = 0.718$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \checkmark$$

$$= 1 + \frac{0.282}{0.718}$$

$$\approx 1.4 \quad (1 \text{ dec. place.}) \checkmark$$

(12)

QUESTION 4.

4.

$$(a) \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{2\pi}{3} - -\frac{\pi}{3}$$

$$= \pi \quad \checkmark$$

(b)

$$P(x) = 2x^3 + ax^2 - 13x + b$$

$$x^2 - x - 6 = (x-3)(x+2)$$

$(x+2)$ is a factor: $P(-2) = 0$

$(x-3)$ is a factor: $P(3) = 0$.

$$P(-2): -16 + 4a + 26 + b = 0 \quad (1)$$

$$P(3): 54 + 9a - 39 + b = 0 \quad (2)$$

$$4a + b = -10 \quad (1)$$

$$9a + b = -15 \quad (2)$$

$$(2) - (1): 5a = -5$$

$$a = -1 \quad \checkmark$$

$$b = -6 \quad \checkmark$$

(c) (i)

$$\text{Let } \cos x - \sqrt{3} \sin x \equiv R \cos(x+\alpha)$$

$$\cos x - \sqrt{3} \sin x \equiv R \cos \alpha \cos x - R \sin \alpha \sin x$$

equating co-efficients:

$$\cos x: R \cos \alpha = 1 \quad (1)$$

$$\sin x: R \sin \alpha = \sqrt{3} \quad (2)$$

$$\frac{(2)}{(1)}: \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \quad \checkmark$$

$$(1)^2 + (2)^2: R^2 = 4$$

$$R = 2 \quad \checkmark$$

$$\cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$$

$$(ii) \cos x - \sqrt{3} \sin x = 1$$

$$2 \cos\left(x + \frac{\pi}{3}\right) = 1$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = 2\pi n + \frac{\pi}{3} \quad \text{or} \quad 2\pi n - \frac{\pi}{3}$$

$$x = 2\pi n \quad \text{or} \quad x = 2\pi n - \frac{2\pi}{3} \quad \checkmark$$

$$(d)(i) T = 20 + Ae^{Kt}$$

$$\frac{dT}{dt} = KAe^{Kt}$$

$$= K(T-20)$$

It satisfies the DE. $\checkmark$

$$(ii) \text{ When } t=0, T=80^\circ\text{C}$$

$$80 = 20 + A$$

$$A = 60 \quad \checkmark$$

$$T = 20 + 60e^{Kt}$$

$$\text{When } t=15, T=40^\circ\text{C}$$

$$40 = 20 + 60e^{15K}$$

$$e^{15K} = \frac{1}{3}$$

$$15K = \ln \frac{1}{3} \quad \checkmark$$

$$15K = -\ln 3$$

$$K = -\frac{1}{15} \ln 3 \quad \text{as required.}$$

$$(iii) T = 20 + 60e^{-\frac{30 \ln 3}{15} t}$$

$$= 20 + 60e^{-2 \ln 3}$$

$$= 20 + 60e^{\ln \frac{1}{9}}$$

$$= 20 + 60 \times \frac{1}{9}$$

$$= 26\frac{2}{3}^\circ\text{C} \quad \checkmark$$

12

QUESTION 5

5.

(i) $x = 2ap$

$y = ap^2$

$\frac{dx}{dp} = 2a$

$\frac{dy}{dp} = 2ap$

$\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx}$

$= 2ap \times \frac{1}{2a}$

$= p$

gradient at $P = p$, $(2ap, ap^2)$

$y - y_1 = m(x - x_1)$

$y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$ is

the equation of the tangent at P.

(ii)

Gradient of PS = Gradient of QP

$\frac{ap^2 - a}{2ap} = \frac{ap^2 - aq^2}{2ap - 2aq}$

$\frac{p^2 - 1}{2p} = \frac{a(p - q)(p + q)}{2a(p - q)}$

$p^2 - 1 = p(p + q)$

$p^2 - 1 = p^2 + pq$

$\therefore pq = -1$ as required.

(iii)

Let m_1 be the gradient of PQ and m_2 be the gradient of PT.

$m_1 = \frac{p+q}{2}$ $m_2 = p$

$\tan \angle TPQ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{\frac{p+q}{2} - p}{1 + p \frac{p+q}{2}} \right|$

$= \left| \frac{p+q - 2p}{2 + p^2 + pq} \right|$ ($pq = -1$)

$= \left| \frac{q - p}{p^2 + 1} \right|$ ($p = -\frac{1}{q}$)

$= \left| \frac{q + \frac{1}{q}}{\frac{1}{q^2} + 1} \right|$

$= \left| \frac{q^3 + q}{1 + q^2} \right|$

$= \left| \frac{q(q^2 + 1)}{(q^2 + 1)} \right|$

$= |q|$

(b) (i)

$\ddot{x} = -n^2 x$

$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -n^2 x$

$\frac{1}{2} v^2 = -\frac{n^2 x^2}{2} + c$

When $x = a$, $v = 0$ and $c = \frac{n^2 a^2}{2}$

so $v^2 = -n^2 x^2 + n^2 a^2$

$v^2 = n^2 (a^2 - x^2)$

(ii) $v^2 = n^2 (a^2 - x^2)$

When $x = 3$, $v = 8$

$64 = n^2 (a^2 - 9)$ — (1)

When $x = 4$, $v = 6$

$36 = n^2 (a^2 - 16)$ — (2)

(1) : $\frac{a^2 - 9}{a^2 - 16} = \frac{64}{36}$

(2) : $\frac{a^2 - 9}{a^2 - 16} = \frac{64}{36}$

Q5 continued.

$$9(a^2 - 9) = 16(a^2 - 16)$$

$$9a^2 - 81 = 16a^2 - 256$$

$$7a^2 = 175$$

$$a^2 = 25$$

$$a = 5 \text{ (amplitude } > 0)$$

$$\textcircled{2} \quad 6 = n\sqrt{25-16}$$

$$3n = 6$$

$$n = 2.$$

$$T = \frac{2\pi}{n}$$

$$= \frac{2\pi}{2}$$

$$T = \pi \text{ s}$$

(ii) The maximum acceleration occurs when $x = a$.

$$\ddot{x} = -n^2 x$$

$$\ddot{x} = -4 \times 5$$

(12)

maximum acceleration is

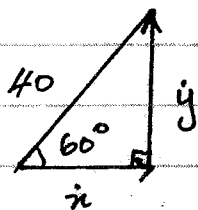
$$20 \text{ m/s}^2$$

(negative implies direction.)

QUESTION 6.

7.

(a)(i)



when $t=0$.

$$\begin{aligned} \dot{x} &= 40 \cos 60^\circ & \dot{y} &= 40 \sin 60^\circ \\ \dot{x} &= 20 \text{ m/s} & \dot{y} &= 20\sqrt{3} \text{ m/s} \end{aligned}$$

$$\ddot{y} = -10 \quad \ddot{x} = 0$$

integrate w.r.t t .

$$\dot{y} = -10t + c_1 \quad \dot{x} = c_2$$

when $t=0$, $\dot{y} = 20\sqrt{3}$ and $\dot{x} = 20$

thus $c_1 = 20\sqrt{3}$ and $c_2 = 20$.

$$\dot{y} = 20\sqrt{3} - 10t \quad \dot{x} = 20 \quad \checkmark$$

integrate w.r.t t .

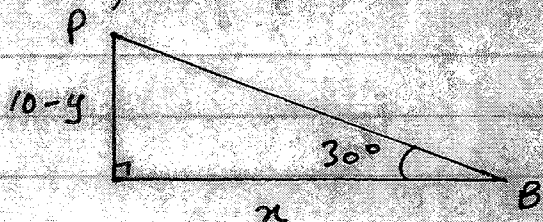
$$y = 20t\sqrt{3} - 5t^2 + c_3 \quad x = 20t + c_4$$

when $t=0$, $y = 10$, $x = 0$.

thus $c_3 = 10$ and $c_4 = 0$

The equations of motion $\checkmark$
are $y = 20t\sqrt{3} - 5t^2 + 10$
 $x = 20t$.

(ii) When the ball hits the plane at P:



$$\frac{10-y}{x} = \tan 30^\circ \quad \checkmark$$

$$\frac{5t^2 - 20t\sqrt{3}}{20t} = \frac{1}{\sqrt{3}} \quad \checkmark$$

$$\frac{t - 4\sqrt{3}}{4} = \frac{1}{\sqrt{3}}$$

$$t - 4\sqrt{3} = \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

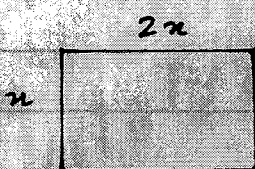
$$t = 4\sqrt{3} + \frac{4\sqrt{3}}{3}$$

$$t = \frac{16\sqrt{3}}{3} \text{ s.} \quad \checkmark$$

(b) $\angle TAO = 90^\circ$ (TA is a tangent)
 $\angle TDO = 90^\circ$ (A line drawn from the centre to the mid-point of the chord is perpendicular to the chord.)

TAOD is a cyclic quadrilateral since opposite angles are supplementary.
Hence $\angle AOT = \angle ADT$ (angles standing on the same arc - drawn to the circumference are equal.) $\checkmark$

(c)



let P = perimeter
let A = area

$$A = 2x^2$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt} \quad \text{by chain rule}$$

$$\frac{dA}{dt} = 4x \times \frac{dx}{dt}$$

$$\text{at } x = 100 \text{ cm.} \quad 18 = 400 \times \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{9}{200} \quad \checkmark$$

$$\text{Now } P = 6x \quad \checkmark$$

$$\frac{dP}{dt} = 6 \times \frac{dx}{dt}$$

$$= 6 \times \frac{9}{200}$$

$$= \frac{27}{100} \text{ cm/s.} \quad \checkmark$$

(12)

QUESTION 7

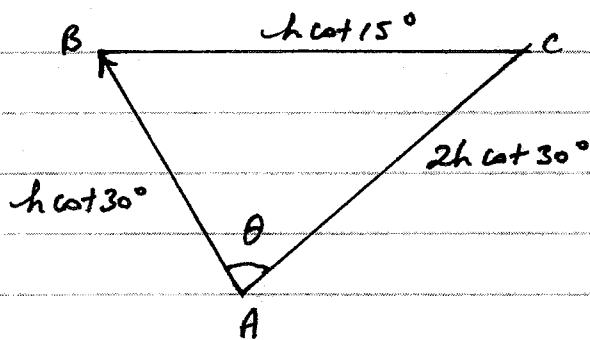
(a) $\triangle DEF: \tan 15^\circ = \frac{h}{DF}$

$$DF = h \cot 15^\circ$$

and $BC = h \cot 15^\circ$. ✓

$$\triangle ABD: AB = h \cot 30^\circ$$

$$\triangle AEC: AC = 2h \cot 30^\circ$$
 ✓



$\triangle ABC$, using the cosine rule:

$$\cos \theta = \frac{h^2 \cot^2 30^\circ + 4h^2 \cot^2 30^\circ - h^2 \cot^2 15^\circ}{4h^2 \cot^2 30^\circ}$$
 ✓

$$= \frac{5 \cot^2 30^\circ - \cot^2 15^\circ}{4 \cot^2 30^\circ}$$

$$= \frac{15 - \cot^2 15^\circ}{12}$$

$$\cos \theta = 0.089316$$

$$\theta = 84^\circ 53'$$

The bearing of the taller tower from A is N $84^\circ 53'$ E. ✓

QUESTION 7 Continued.

9.

$$(b) (1+x)^{n-1} = \binom{n-1}{0} + \binom{n-1}{1}x + \binom{n-1}{2}x^2 + \dots + \binom{n-1}{n-1}x^{n-1} \quad \checkmark$$

integrating both sides w.r.t x :

$$\frac{(1+x)^n}{n} = \binom{n-1}{0}x + \frac{1}{2}\binom{n-1}{1}x^2 + \frac{1}{3}\binom{n-1}{2}x^3 + \dots + \frac{1}{n}\binom{n-1}{n-1}x^n + C_1$$

put $x=0$.

$$\frac{1}{n} = C_1$$

$$\frac{(1+x)^n}{n} = \binom{n-1}{0}x + \frac{1}{2}\binom{n-1}{1}x^2 + \frac{1}{3}\binom{n-1}{2}x^3 + \dots + \frac{1}{n}\binom{n-1}{n-1}x^n + \frac{1}{n} \quad \checkmark$$

put $x=7$.

$$\frac{8^n}{n} = 7\binom{n-1}{0} + \frac{7^2}{2}\binom{n-1}{1} + \frac{7^3}{3}\binom{n-1}{2} + \dots + \frac{7^n}{n}\binom{n-1}{n-1} + \frac{1}{n} \quad \checkmark$$

$$\frac{1}{n}(8^n - 1) = 7\binom{n-1}{0} + \frac{7^2}{2}\binom{n-1}{1} + \frac{7^3}{3}\binom{n-1}{2} + \dots + \frac{7^n}{n}\binom{n-1}{n-1}$$

$$\therefore 7\binom{n-1}{0} + \frac{7^2}{2}\binom{n-1}{1} + \frac{7^3}{3}\binom{n-1}{2} + \dots + \frac{7^n}{n}\binom{n-1}{n-1} = \frac{1}{n}(2^{3n} - 1) \quad \checkmark$$

(c) Show the statement is true for $n=2$.

A 1, 2. Sum of the products = $1 \times 2 = 2$.

$$\begin{aligned} \frac{n}{24}(n-1)(n+1)(3n+2) &= \frac{2}{24}(2-1)(2+1)(6+2) \\ &= \frac{1}{24} \times 1 \times 3 \times 8 \\ &= 2 \quad \checkmark \end{aligned}$$

It's true for $n=2$.

B Assume that the result is true for $n=k$.

ie/ The sum of the products of all the pairs of integers that can be formed from the first k positive integers is $\frac{k}{24}(k-1)(k+1)(3k+2)$.

Prove that it is true for $n=k+1$.

The addition of the integer $(k+1)$ will add another $(1+2+3+\dots+k)(k+1)$ products in pairs.

Q7 (c) continued.

10.

When $n = k+1$ the sum is :

$$\frac{k}{24} (k-1)(k+1)(3k+2) + (k+1)(1+2+3+\dots+k) \checkmark$$

$\left(1+2+3+\dots+k \right.$ is an arithmetic series. $\text{Sum} = \frac{k}{2}(1+k)$

$$= \frac{k}{24} (k-1)(k+1)(3k+2) + (k+1) \frac{k}{2} (k+1) \checkmark$$

$$= \frac{k}{24} (k+1) \{ (k-1)(3k+2) + 12(k+1) \}$$

$$= \frac{k}{24} (k+1) \{ 3k^2 + 11k + 10 \}$$

$$= \frac{k}{24} (k+1)(k+2)(3k+5) \text{ as required. } \checkmark$$

c// It follows from parts A and B by mathematical induction that the statement is true for all positive integers $n \geq 2$.

12